4G Capital × Underwrite for Collection

Merchant cash flows, repeat lending and the economics of risk sharing

A reproducible case connects distributor-financed stock to borrower cash, lender contribution and guarantee settlement.

Proposed first engagement: one product, one defined cohort and one repeat-lending or support decision.

nevilmaloba@philtechent.com

The research questions lead to operating decisions

1. Does each lending cycle leave the merchant able to replenish stock, meet commitments and repay sustainably? The decision is the next amount, due date and evidence required.

2. Which support or authorised adjustment improves merchant cash and durable repayment? The decision is which intervention to offer, when and at what cost.

3. How does risk sharing support additional sustainable lending? The decision combines retained loss, claim settlement, funding requirements and programme eligibility.

4G products and the IFC partnership provide a concrete setting

Public context	Research application
UPIA: 30-day working-capital loans with training and support [2]	Evaluate cash conversion, successive loans and delivered support.
KUZA: 21-day distributor-linked stock finance [3]	Reconcile order, distributor payment, delivery, sale and receipt.
IFC project 53511: unfunded guarantee for a new MSME portfolio [4]	Configure eligibility, layer sharing, claims, recoveries and funding.
IFC advisory focus: underwriting and distributor finance [4]	Connect data definitions and decisions to measurable outcomes.

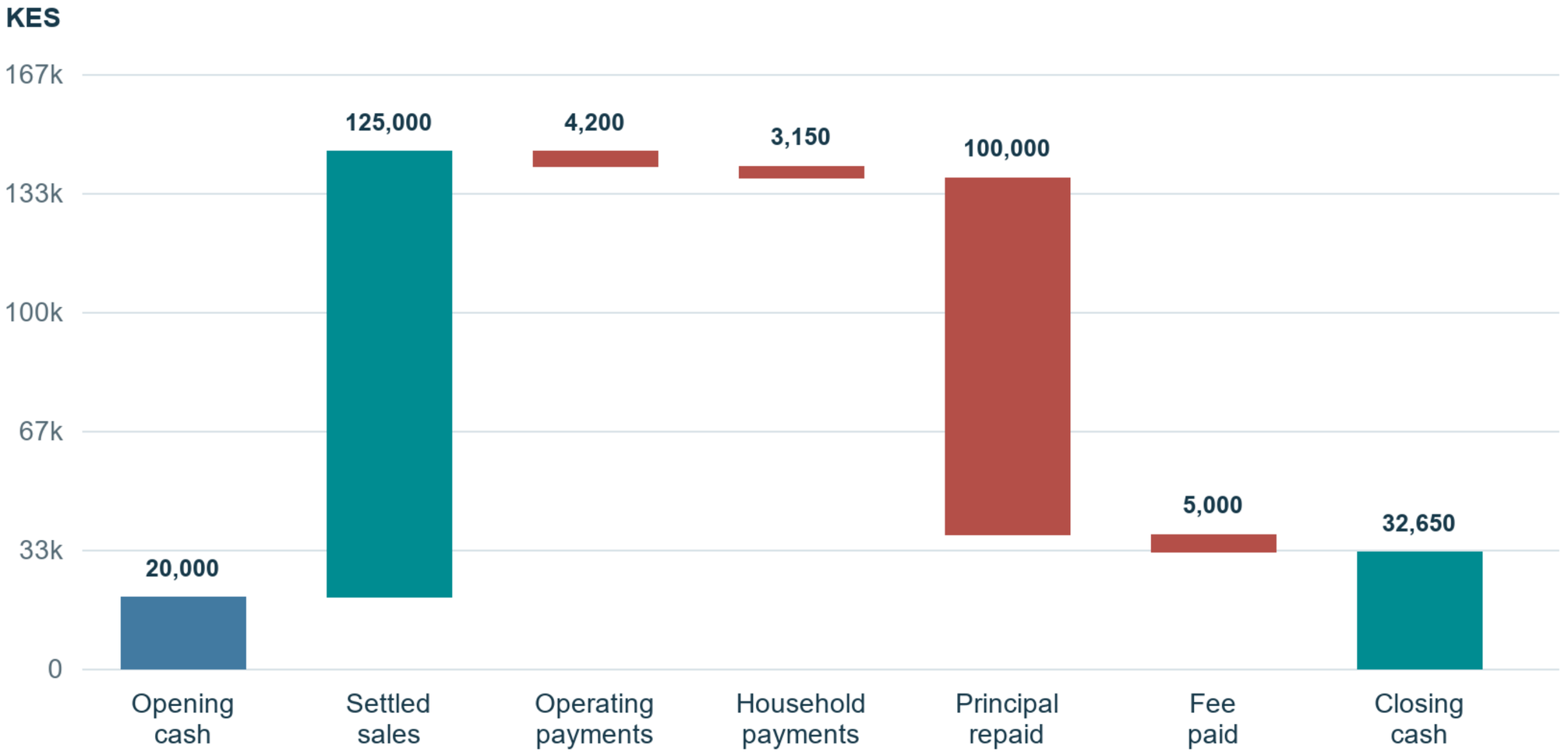
The proposed case gives these workstreams a common borrower, facility, transaction and cash record.

The synthetic case has explicit operating and cost assumptions

Assumption	Configuration
Population and horizon	12 merchants, 180 operating days, runoff through day 300, up to 6 cycles
First facility and merchant cash	KES 100,000 stock finance, KES 20,000 opening cash,21-day tenor
Trading and household needs	25% markup,18 selling days,2-day settlement lag, daily needs KES 200+150
Fees and lender costs	5% fee per cycle,12% annual funding, KES 500 origination, KES 2 per open-loan day
Repeat-lending rule	KES 10,000 retained cash floor, KES 50,000–120,000 limits,7-day cooldown
Guarantee assumptions	1% annual fee, cumulative KES 4.44m inclusion budget, explicit claims and recovery rules

Cash remaining after the first lending cycle

KES 100,000 goes directly to the distributor; it enters inventory, while sales enter merchant cash.



Synthetic KES case | Author-selected assumptions | Daily ledger outputs | 18 October 2026

Repeat lending uses the previous completed operating cycle

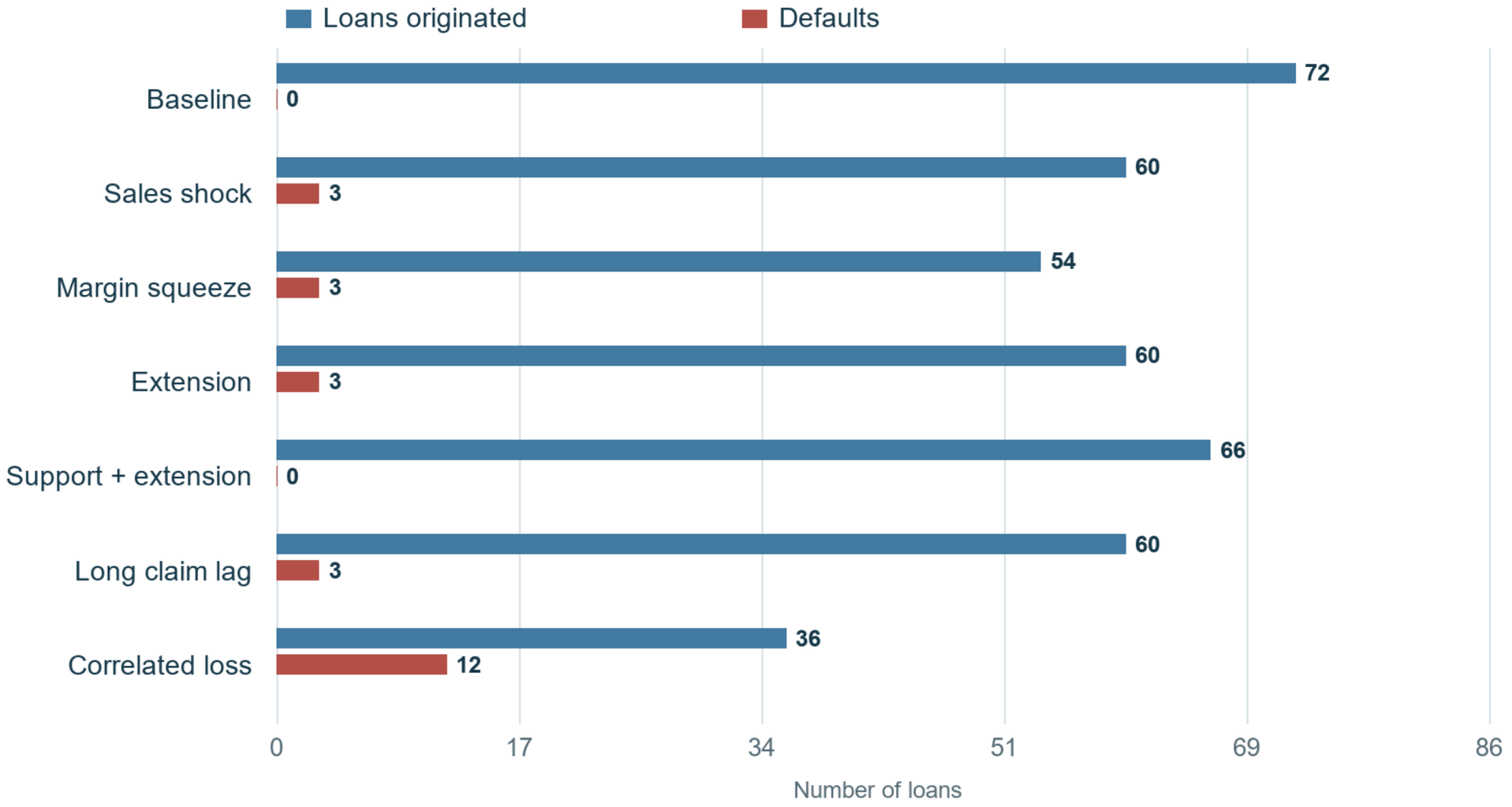
The gate waits for settlement of the previous loan, completion of relevant sales receipts and a seven-day cooling period. The next projection uses the margin, conversion and timing observed by that decision date.

Projected cash after essential expenses and debt service must retain KES 10,000. A timely borrower with sufficient cash can receive a KES 10,000 increment, up to KES 120,000. A weaker projection reduces the amount or defers lending.

The model records each origination and its reasons. A research pilot can compare that rule with approved alternatives while following business cash, repayment, repeat eligibility and customer experience.

Loan originations and defaults under seven scenarios

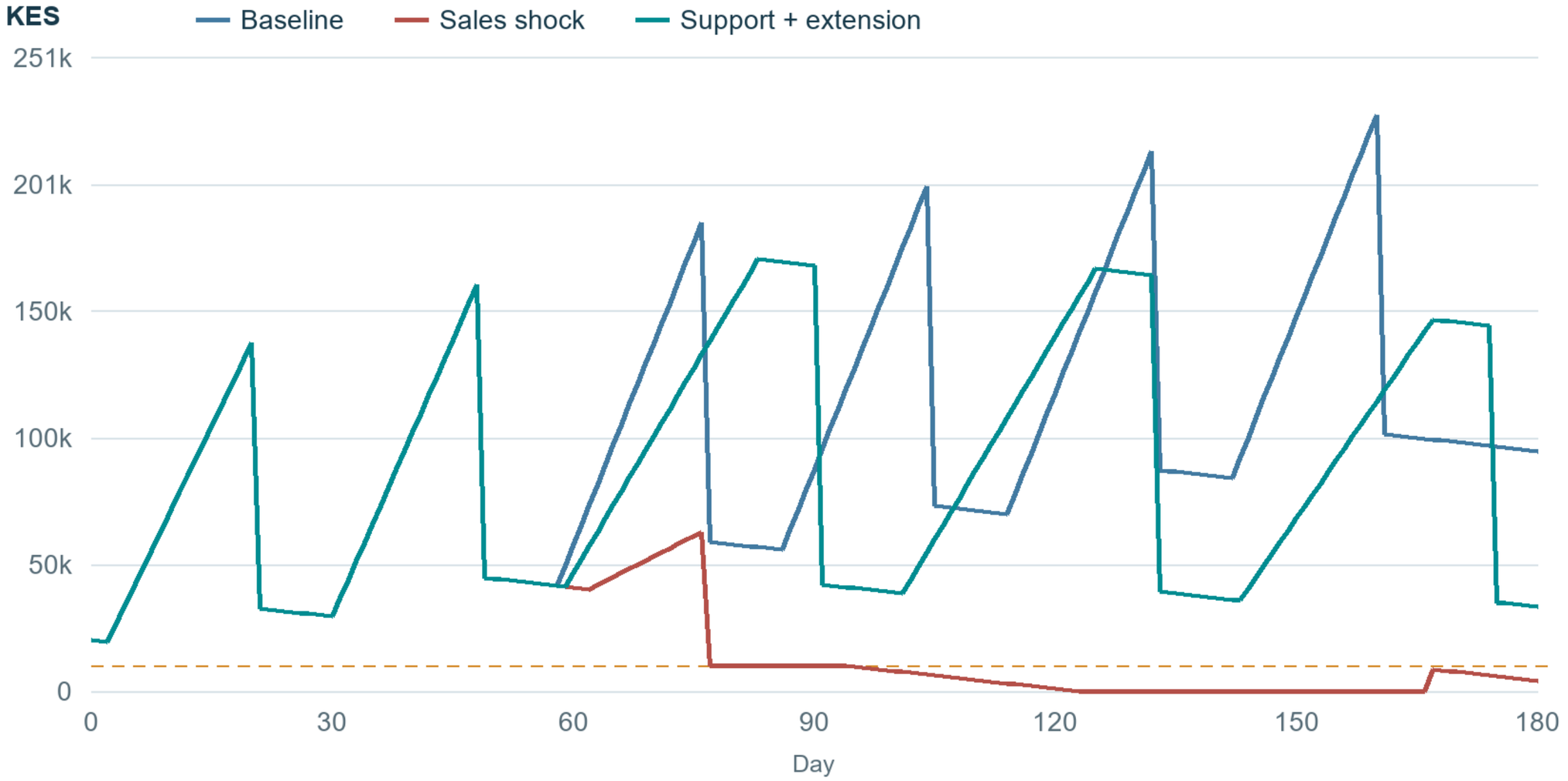
Total originations include each merchant's first loan; defaults count loan events.



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Merchant cash through successive lending cycles

V01 cash through day 180; dashed line marks the KES 10,000 operating floor.



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The support scenario makes a testable economic proposition

Support plus rescheduling produces 66 loans and 0 defaults under the selected trading assumptions. The lender spends KES 21,600 on support. Merchant closing cash totals KES 769,800.

The same scenario carries KES 164,640 of closing inventory at cost. Its valuation and future sale matter alongside cash. Extension alone retains three defaults, so the model separates extra time from an assumed improvement in trading.

A prospective study would estimate the improvement: compare approved support options, record actual delivery and measure cash after commitments, stock conversion, durable repayment and incremental costs.

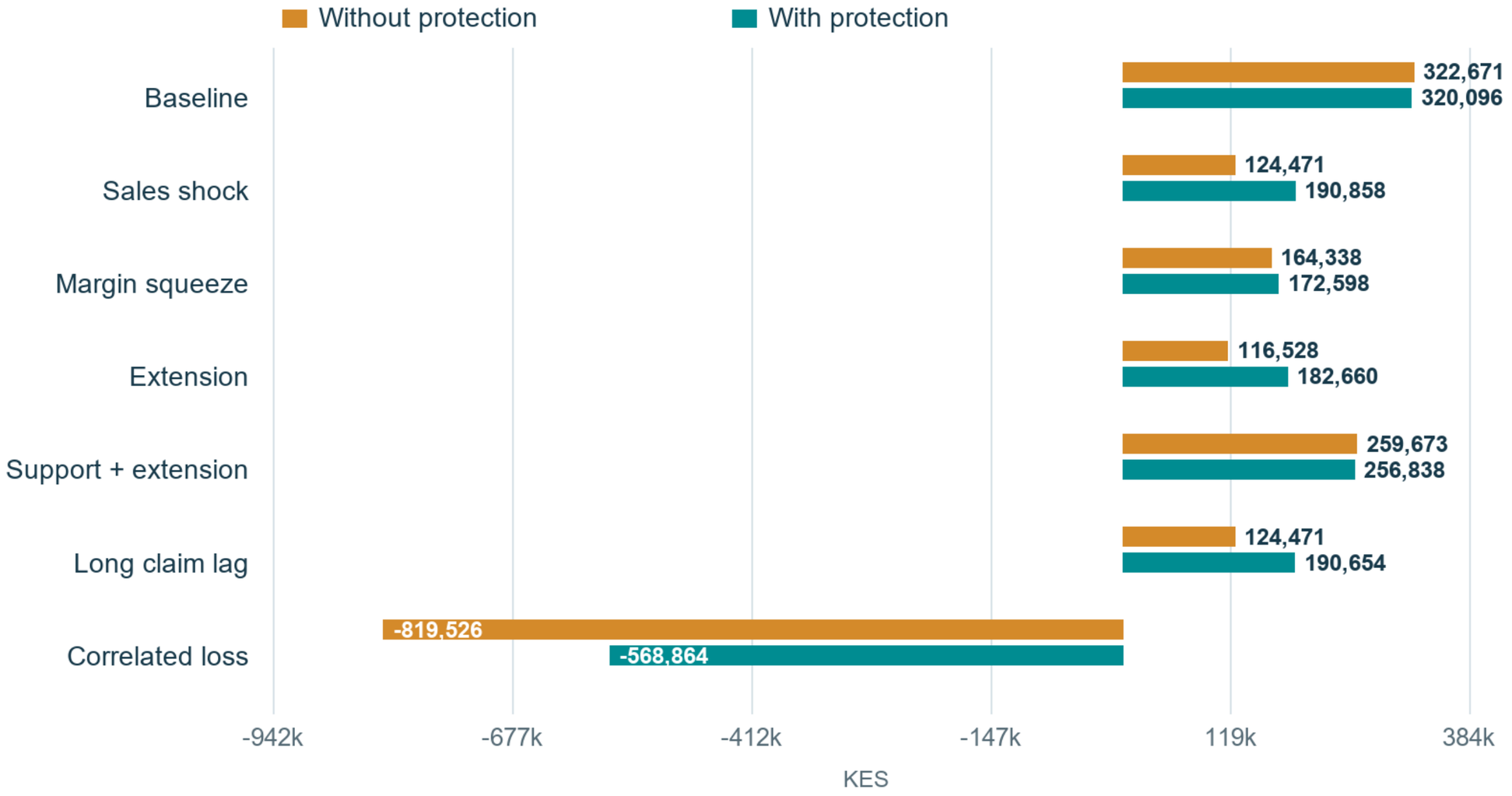
Collected fees reconcile to lender contribution

KES, complete model horizon	Baseline protected	Sales shock protected
Collected fees	414,000	316,500
Origination + servicing + collection + support	39,024	33,936
Funding cost	52,377	44,281
Guarantee fees	2,503	2,634
Retained credit loss after recovery / protection	0	44,791
Realised contribution	320,096	190,858

Contribution precedes tax, central overhead allocation and a capital hurdle. All amounts are synthetic.

Lender contribution under seven trading scenarios

Realised model contribution after funding, operations, losses, recoveries and guarantee cash.



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The comparison identifies distinct commercial decisions

Synthetic result	Economic interpretation
Baseline: KES 320,096 protected vs 322,671 unprotected	With no losses, guarantee fees and their funding cost reduce contribution.
Sales shock: KES 190,858 protected vs 124,471 unprotected	Net claim receipts exceed protection costs. Merchant defaults remain 3 in both paths.
Support + extension: KES 256,838 protected	The assumed trading improvement supports 66 loans. Test that mechanism and its cost locally.
Severe common shock: KES -568,864 protected	Retained losses and funding exceed fee income. Concentration and capital capacity need explicit review.

The teaching guarantee follows two layers and dated eligibility

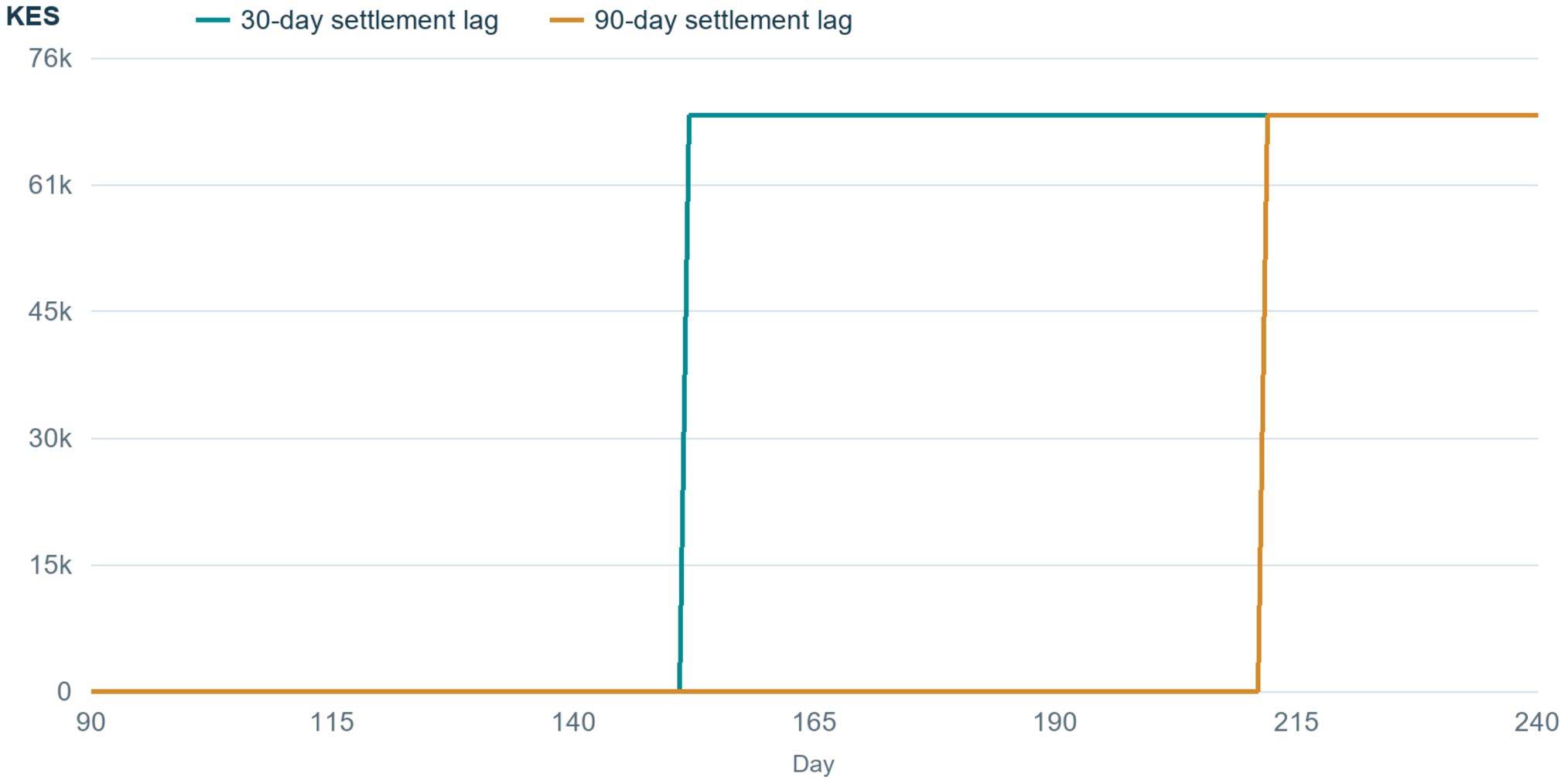
Layer / rule	Synthetic configuration
First layer	KES 220,000 eligible gross loss width, KES 200,000 maximum protection
Second layer	Next KES 4,220,000 width, KES 420,000 maximum protection
Inclusion and recovery	Cumulative originations budget KES 4.44m, explicit ineligible borrower, no capacity reinstatement

$$G(L) = (200/220)\min(L,220{,}000) + (420/4{,}220)\min(\max(L-220{,}000,0),4{,}220{,}000)$$

L is cumulative eligible gross loss. Each new loss receives the increase in G(L).

Guarantee cash under two settlement schedules

Cumulative guarantee receipts less recovery remittances; identical delayed-sales borrower path.



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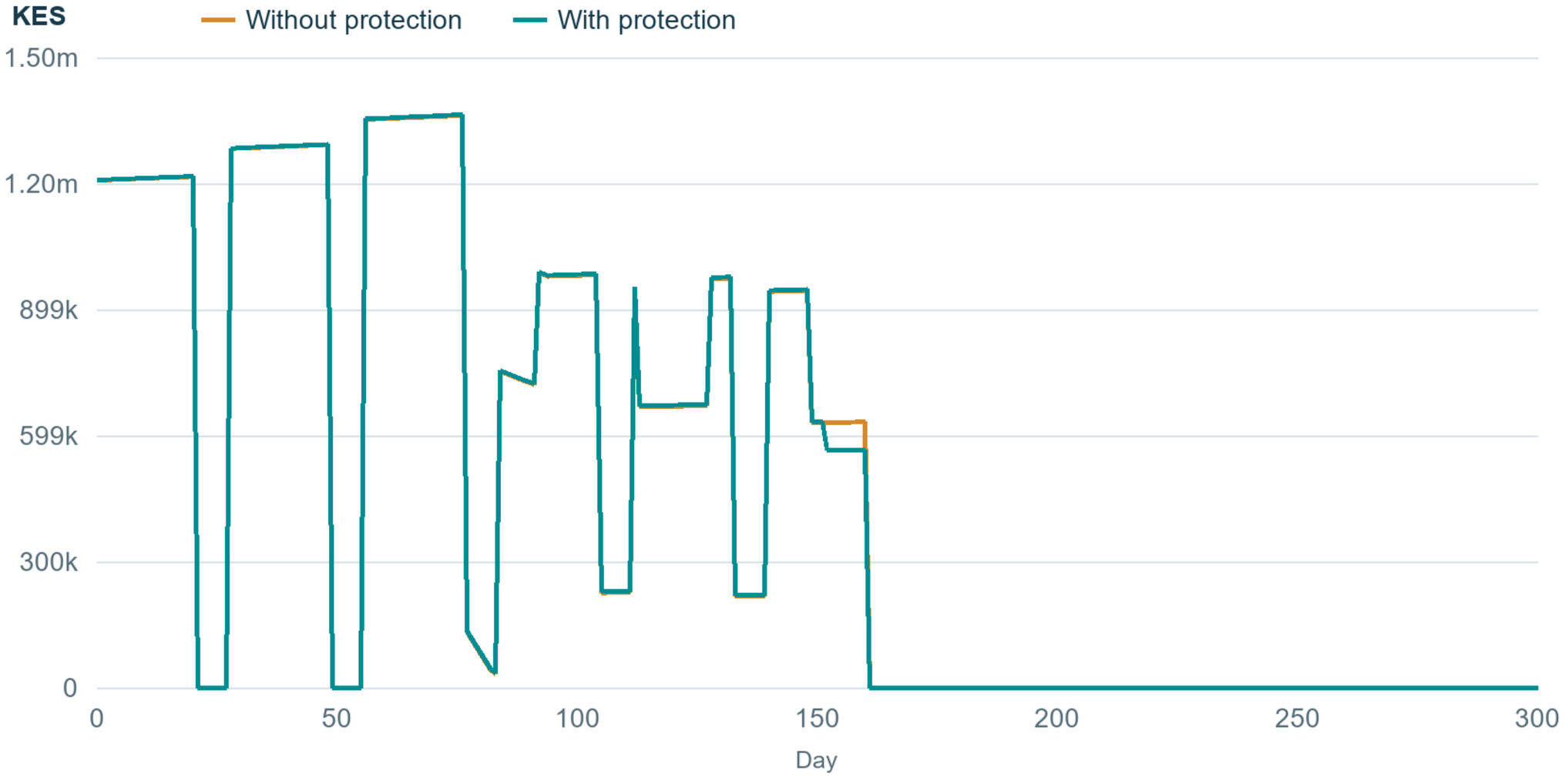
The settlement lag changes funding cost with merchant outcomes fixed

Sales shock, KES	30-day settlement	90-day settlement
Gross written-off principal	113,700	113,700
Borrower recovery	0	0
Guarantee cash received	68,909	68,909
Later recovery remittance	0	0
Retained credit loss	44,791	44,791
Lender contribution	190,858	190,654

The same net protection arrives later. Additional funding reduces contribution by KES 204.

Funding required through lending and claim settlement

Closing borrowing balance; each day reconciles disbursements, collections, costs and protection cash.



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Severe losses expose retained credit risk and merchant strain

Severe common shock, KES	Amount
Gross written-off principal	894,000
Less: borrower recovery	62,400
Less: guarantee cash received	259,666
Add: recovery remitted to guarantor	18,124
Retained credit loss	590,058
Lender contribution after all costs	-568,864

Merchant cash totals KES 65,400. Unpaid household need totals KES 81,000. These outcomes accompany the financial result.

Agentic workflow with controlled calculations and human approval

Step and accountable owner	Evidence or action
1. Intake agent and data steward	Reconcile orders, stock, receipts and loan records. Record purpose, provenance and freshness.
2. Calculation service and model owner	Run cash projections, eligibility and approved models. Return versions, units and uncertainty.
3. Case agent and credit officer	Assemble options and supporting evidence. Route missing data and policy exceptions for review.
4. Maker and independent checker	Approve a limit, due date or material intervention using the permitted action set.
5. Execution service and operations	Send authorised instructions to lending/CRM systems. Capture idempotency keys and receipts.
6. Monitoring agent and finance/risk	Reconcile outcomes, claim events and overrides. Trigger review, then update the evidence record.

A focused pilot links each phase to an acceptance decision

Phase	Working output	Decision to proceed
1. Product and data definition	One product/cohort, evidence dictionary, permissions and executed protection terms	Credit, operations, legal and finance agree the scope.
2. Historical reconstruction	Reconciled cash cycles, claims history and time-based validation	Independent reviewer reproduces balances and results.
3. Shadow recommendations	Proposed limits/support with uncertainty and officer feedback	Model owner confirms usefulness, stability and review capacity.
4. Approved prospective study	Assigned support, delivered actions, costs and customer/lender outcomes	Evaluate prespecified outcomes, uncertainty and operational effects.

Data access and decision rights enter the specification

The lender keeps identifiable customer data in its approved environment. Purpose-bound access, pseudonymised research views, retention rules and transfer arrangements become explicit requirements.

A complete decision packet records business and facility IDs, relevant transactions, decision time, policy version, available actions and authorised reviewers. CRM and the lending subledger reconcile the resulting action.

Customer-facing changes follow the approved review route. The implementation records notice, meaningful explanation, challenge and human review under the applicable Kenyan data-protection framework.

The proposed engagement produces a usable decision pack

Merchant evidence: reconciled stock-to-cash cycles, repeat eligibility, operating obligations and borrower outcomes. Each measure has a defined period and economic meaning.

Lender evidence: collected revenue, servicing/support costs, gross and retained losses, claim timing, liquidity demand and concentration stress. These connect pricing and growth decisions.

Proposed next step: a technical scoping meeting with product, credit, operations, finance and data teams to select the first cohort and agree the acceptance criteria.

Nevil Maloba, Philtechent Limited nevilmaloba@philtechent.com

Sources and reproducible material

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Code: github.com/TheAlchemistNerd/mwendo-pamoja-continuous-underwriting
Folder: research_lab/four_g_merchant_finance_case